

$$Q = \sqrt{(920^2 + (500 + 1302)^2)} = 2023 \text{ N} \sim 2000 \text{ N}$$

2 (a) By C.O.E.

$$KE_A + GPE_A = WD_{Friction} + KE_B + GPE_B$$

$$\frac{1}{2}(300)(5.0^2) + 35kJ = 15kJ + \frac{1}{2}(300)(v_B^2)$$

$$= 12.6 \text{ m s}^{-1}$$

(b) From N2L, $F = ma$ and WD by a constant force F , $W = F.s = mas$

From equations of motion, $v^2 = u^2 + 2as$

$$as = \frac{1}{2}(v^2 - u^2)$$

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \text{ (final KE - initial KE)}$$

Assuming body starts from rest, i.e. $\frac{1}{2}mu^2 = 0$

$$W = \frac{1}{2}mv^2 \text{ (derived)}$$

Assumption:

Constant force F

No energy loss / smooth surface

No initial KE / starts from rest